CEBU INSTITUTE OF TECHNOLOGY UNIVERSITY

COLLEGE OF COMPUTER STUDIES

Software Project Proposal

Scribbie: An English Learning App for Grade 1 Students of Cebu Institute of Technology

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Executive Summary

This project aims to address a critical challenge faced by Grade 1 students and teachers at Cebu Institute of Technology – University (CIT-U) which is the student's disinterest and lack of motivation to develop basic writing and reading skills. Research shows that students often struggle with literacy due to short attention spans, lack of engagement, and frustration with traditional teaching methods (Rosmayasari, 2021). The current curriculum's reliance on textbook

based practices fails to capture the curiosity and creativity of children aged 6–7, leading to disengagement, poor focus, and resistance to learning. As a result, students miss essential opportunities for skill development, and their overall performance suffers (Persada & Lutfi, 2020). Teachers, in turn, face difficulties maintaining classroom interest and managing behavior without the aid of interactive and stimulating learning materials (Surendeleg et al., 2014).

To address these issues, we propose a gamified software solution designed to enhance the existing curriculum with engaging lessons, interactive quizzes, and creative writing exercises. It's vital to catch the attention of students since the role of attention plays substantially in the development of writing fluency and quality of younger students (Kent et al., 2013, p. 1165). Gamification in education has been proven to increase student motivation and learning outcomes by making educational activities more interactive and enjoyable (Estigarribia & Clark, 2007). Through this platform, writing becomes a fun and accessible endeavor for students while equipping teachers with user-friendly tools to monitor progress and tailor lessons to individual needs. This approach not only improves student engagement and enhances writing proficiency but also helps create a more dynamic and manageable classroom environment.

By bridging the gap between traditional instruction and the needs of modern learners, this project aims to lay a strong foundation for literacy development. It empowers students to cultivate essential reading and writing skills while providing educators with innovative resources to foster creativity and active participation, ultimately shaping a more effective and enriching learning experience for all.

Background and Problem Statement

Grade 1 students often struggle with basic writing skills due to a lack of interest, short attention spans, and frustration with traditional teaching methods. Research indicates that students with learning difficulties often exhibit lower academic self-esteem and struggle with basic literacy skills, including reading and writing, which significantly impacts their overall educational development (Rosmayasari, 2021). The current curriculum relies heavily on textbooks and static materials, which fail to engage young learners effectively. This results in disengagement, reluctance to learn, and missed opportunities for skill development (Persada & Lutfi, 2020). Teachers also face challenges in maintaining student interest and managing classroom behavior due to the lack of interactive and visually engaging materials.

English teachers at Cebu Institute of Technology – University face similar challenges in maintaining student interest and managing classroom behavior. Without engaging resources, they are unable to create a stimulating learning environment that encourages the student's creativity and active participation. The lack of interactive tools results in passive learning, which has been shown to limit student motivation and engagement (Surendeleg et al., 2014). This gap in the curriculum not only impacts students' immediate learning but also has long-term consequences for their literacy development and confidence in writing.

Existing solutions, such as remedial classes and additional teaching hours, are insufficient to address the root causes of disengagement. These methods do not incorporate interactive or gamified elements, which are essential for capturing the attention of young learners. Research on gamification in education demonstrates that integrating game-like elements into learning environments can significantly increase student engagement and motivation by making learning more interactive and enjoyable (Estigarribia & Clark, 2007). Furthermore, teachers often lack the resources and training to implement innovative teaching strategies effectively, leading to a reliance on outdated methods that fail to address students' diverse learning needs (Persada & Lutfi, 2020).

The existing curriculum relies solely on textbooks and static materials, which are not designed to engage young learners effectively. There are no interactive or gamified tools to make writing fun and relatable for students. This results in disengagement, reluctance to learn, and missed opportunities for skill development (Rosmayasari, 2021). Additionally, teachers lack the resources to monitor student progress effectively or tailor lessons to individual needs, further exacerbating the problem (Surendeleg et al., 2014).

A gamified software solution is necessary to address the limitations of the current curriculum. By incorporating fun, interactive lessons, quizzes, and creative activities, the proposed solution will:

- Capture students' attention and maintain their interest through engaging and fun activities
- Foster creativity and real-world application of writing skills, making learning more meaningful
- Provide teachers with tools to monitor student progress, customize lessons, and create a more dynamic and effective learning environment

This solution aligns with the needs of modern learners, who are increasingly drawn to interactive and technology-driven learning methods. It also supports teachers in delivering more effective instruction, ultimately benefiting both students and educators.

Project Objectives

The goal is to maximize students' engagement, enhance reading and writing capability, and aid the teachers with an integrated learning platform. In the initial half-year of deployment, the goal is to achieve a 40% boost in students' engagement in writing and reading exercises. Also, within one school year, the goal is to enhance students' proficiency in letter, word, and sentence creation by 30%. For helping the teachers, the platform will offer a simple and intuitive interface for tracking students' progress and interactive lessons, and this capability shall be fully implemented during development.

Functional Requirements:

• Interactive lessons with animations, storytelling, and step-by-step guides. • Gamified activities such as word puzzles, theme matching, and sentence-building games.
• Quizzes with instant feedback and progress tracking for students.
• Teacher dashboard for monitoring student performance and customizing lessons.

Non-Functional Requirements:

• User-friendly interface designed especially for young learners and teachers. • Eye catching graphics to easily catch student's attention
• Scalable and secure platform to protect student data and ensure accessibility. • Compatibility with computers for easy access at the comfort of the student's home • Responsive website considering frequent network connection issues.

Documentation:

• Software Requirements Specification (SRS) detailing system requirements. • Software Design Description (SDD) outlining architecture, modules, and design decisions.
• Software Project Management Plan (SPMP) including project timeline, milestones, and resources.
• Software Test Document with test cases, scenarios, and results.

Usability Testing and User Survey:

1. Conduct usability testing with Grade 1 students and teachers to evaluate the system's effectiveness.
2. Collect feedback to refine and improve the system before full deployment.

Final Paper (ACM Format):

1. Summarize the project's results and discussions in ACM format, ensuring compliance with academic and industry standards.

Scope and Limitations

• Interactive reading and writing lessons
• Eye-catching graphics that easily grasp the children's attention
• Gamified activities such as word puzzles, sentence-building games, and creative writing prompts with assistance of AI.
• Progress tracking and reporting tools for teachers.
• Real-world application scenarios to make writing more relatable.
• Limited to Grade 1 students and English writing skills.
• Development timeline of 8 weeks, with a focus on core functionalities. • Budget constraints may limit the inclusion of premium features from AI services

Proposed Solution and Methodology

Overview of the Software

With the consideration that students are not allowed to bring gadgets on the campus, the application will serve as a study material and assignment for the students to take at home. The software will also have a gamified learning experience with interactive lessons, quizzes, and creative activity to make writing and reading engaging for Grade 1 students, as well as supplement the resources that will be provided by teachers. The Teachers will
also receive a dashboard to monitor progress, customize lessons, and support instruction. The platform will also be accessible for use on computers and offer convenience for both the class and home setting.

Technologies and Platforms

- Frontend: HTML5, CSS3, React.JS
- Backend: Springboot
- Database: MongoDB for storing student progress and lesson data.
- Gamification Tools: To be discussed
- Deployment: Cloud-based platform for scalability and accessibility.

Development Approach

- Agile Methodology: Iterative development with regular feedback from teachers and students.
- Sprints: Two-week sprints to deliver incremental features and ensure continuous improvement.
- Testing: Continuous testing and refinement based on user feedback to ensure the system meets the needs of its users.

Target Users, Customers, Beneficiaries, and Partners

Intended Users

- Grade 1 students (aged 6-7) at CIT-U's Elementary Department.
- English elementary teachers at CIT-U's Elementary Department.

How the Solution Benefits Them

- Students: Engaging and fun way to learn writing skills, fostering creativity and real world application.
- Teachers: Tools to monitor progress, customize lessons, and create a more dynamic and manageable learning environment.

Potential Partners or Stakeholders

- CIT-U's Elementary Department for pilot testing and implementation.

Technical Requirements

Hardware and Software Needs

- Computer and Internet Connection for students and teachers to access the platform.
- Cloud infrastructure for hosting the platform and ensuring scalability.

Security and Infrastructure Considerations

- Data encryption to protect student information and ensure privacy.
- Scalable cloud infrastructure to handle multiple users and ensure smooth performance.

Evaluation and Success Metrics:

Key Performance Indicators (KPIs)

- Student Engagement Rates: Measured by participation in activities and time spent on the platform.
- Improvement in Writing Skills: Assessed through quizzes, teacher evaluations, and pre- and post-deployment assessments.
- Teacher Satisfaction: Measured through surveys and feedback on the platform's usability and effectiveness.

Testing and Validation Strategies

- Usability Testing: Conducted with Grade 1 students and teachers to evaluate the system's effectiveness and identify areas for improvement.
- Pilot Testing: Implemented in select sections and function as a material for the student's assignment before full deployment to gather real-world feedback and refine the system.

Conclusion

The goal of this project is to develop a gamified computer program designed to address the challenges faced by Grade 1 teachers and students in teaching and learning basic writing skills. Research has shown that early literacy development is heavily influenced by students' attention, language skills, and instructional quality (Kent et al., 2013). However, the current reliance on textbook-based, static teaching methods often leads to disengagement and frustration among young learners (Rosmayasari, 2021). By incorporating interactive and enjoyable learning materials, this program aims to increase students' engagement, enhance their writing fluency, and foster a more effective and motivating learning environment.

The proposed system will not only provide students with an accessible and entertaining platform for learning basic writing but will also empower teachers with tools to deliver high quality instruction and track student progress. Research on gamification in education has demonstrated its potential to enhance motivation and learning outcomes by making educational experiences more interactive and enjoyable (Estigarribia & Clark, 2007). Furthermore, the integration of technology-driven approaches can help address the lack of time and resources often allocated to writing instruction in early education (Kent et al., 2013).

We strongly endorse the adoption and implementation of this innovative solution. By bridging the gap between traditional instructional methods and the evolving needs of modern learners, this project has the potential to make a significant contribution to the development of literacy skills and the strengthening of learning dynamics at CIT-U's elementary department. Through this approach, we aim to foster a learning environment that inspires creativity, builds confidence in writing, and supports long-term academic success.

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